

TASK 2

Suspicious Behavior Investigation Report

Rev D

Task 2

SUSPICIOUS BEHAVIOR INVESTIGATION

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2 Project background and description

2.1 Possible Insider Threat?

Background

A report was received that an IT support employee had weird text on his screen that didn't seem to be work related. It is not known if this is serious but does need to be investigated. Network traffic from that time period has been captured.

Download the packet capture (PCAP) file and using NetworkMiner and WireShark analyze what the user was doing and what happened.

- Is this an insider threat data breach?
- Does incident response need to be brought in?

These findings will determine if law enforcement or other actions need to be taken.

3 Executive Summary

The employee under investigation first used Google to search for fire tornadoes. They then downloaded documentation pertaining to a Mozilla file server before visiting a game site at zeemud.org.

The weird text seen is a cover page for the game site Zee Mud (shown in the screen grabs).

Currently, we have no evidence of any malicious activities. However, we do wonder if this employee might have violated the corporate Acceptable Use Policy

3.1 Recommendations:

- Ensure that there is an up-to-date Acceptable User Policy (AUP) in place.
 - Is the AUP easily accessible to ALL employees and do they understand it?
- Conduct periodic AUP employee updates and reviews.
 - Is this employee in compliance with the current AUP?
- Conduct an annual risk assessment of the AUP to ensure it remains up-to-date and post for all employees to review.

4 Technical Appendix

4.1 Employee IP address

10.101.1.56

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4.2 NetworkMiner Screen Grabs

4.2.1 DNS Tab Screen Grab

Hosts (18) Files (3) Images Messages Credentials (3) Sessions (9) DNS (19) Parameters (51) Keywords Anomalies												
Filter keyword:												
☐ Case sensitive												
Frame nr.	Timestamp	Client	Client Port	Server	Server Port	IP TTL	DNS TTL (time)	Transaction ID	Type	DNS Query	DNS Answer	Alexa Top 1M
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:00:56	0x5B0A	0x0005 (CNAME)	clients1.google.com	clients1.google.com	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.101	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.97	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.102	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.105	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.96	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.103	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.98	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.104	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.110	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.99	N/A (Pro version only)
35	2012-08-15 01:06:28 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:02:45	0x5B0A	0x0001 (A)	clients1.google.com	74.125.225.100	N/A (Pro version only)
45	2012-08-15 01:06:30 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:05:00	0x5908	0x0005 (CNAME)	ftp.download.com	phx1-sha-redirect-lb.cnet.com	N/A (Pro version only)
46	2012-08-15 01:06:30 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:01:26	0x5908	0x0001 (A)	phx1-sha-redirect-lb.cnet.com	64.30.224.118	N/A (Pro version only)
58	2012-08-15 01:06:35 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:03:59	0x5909	0x0005 (CNAME)	ftp.mozilla.org	ftp.dyndect.mozilla.net	N/A (Pro version only)
58	2012-08-15 01:06:35 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:00:30	0x5909	0x0005 (CNAME)	ftp.dyndect.mozilla.net	ftp2-zlb.vips.scl3.mozilla.com	N/A (Pro version only)
58	2012-08-15 01:06:35 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	00:05:00	0x5909	0x0001 (A)	ftp2-zlb.vips.scl3.mozilla.com	63.245.215.56	N/A (Pro version only)
123	2012-08-15 01:06:47 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	02:00:00	0x8CA6	0x0005 (CNAME)	mud.zeemud.org	zeemud.org	N/A (Pro version only)
123	2012-08-15 01:06:47 UTC	10.101.1.56 (Windows)	1305	75.75.76.76	53	53	02:00:00	0x8CA6	0x0001 (A)	zeemud.org	209.128.72.218	N/A (Pro version only)

Green - Google search
Red - Mozilla downloads
Blue - Zeemud game

4.2.2 Host Tab Screen Grab

10.101.1.56 (Windows)	
IP: 10.101.1.56	
MAC: 000C29197E7E	
NIC Vendor: VMware, Inc.	
MAC Age: 2003-01-21	
Hostname:	
OS: Windows	
TTL: 128 (distance: 0)	
Open TCP Ports: 5001 5002	
Sent: 133 packets (8,650 Bytes), 0.00 % cleartext (0 of 0 Bytes)	
Received: 141 packets (24,083 Bytes), 0.00 % cleartext (0 of 0 Bytes)	
Incoming sessions: 2	
Server: 10.101.1.56 (Windows) TCP 5001	
Server: 10.101.1.56 (Windows) TCP 5001 (0 data bytes sent), Client: 63.245.215.56 [ftp2-zlb.vips.scl3.mozilla.com] [ftp.dyndect.mozilla.net] (Linux) TCP 7011 (528 data bytes sent), Session start: 2012-08-15 01:06:44 UTC, Session end: 2012-08-15 01:06:44 UTC	
Server: 10.101.1.56 (Windows) TCP 5002	
Server: 10.101.1.56 (Windows) TCP 5002 (0 data bytes sent), Client: 63.245.215.56 [ftp2-zlb.vips.scl3.mozilla.com] [ftp.dyndect.mozilla.net] (Linux) TCP 7011 (528 data bytes sent), Session start: 2012-08-15 01:06:51 UTC, Session end: 2012-08-15 01:06:52 UTC	
Outgoing sessions: 4	
Server: 63.245.215.56 [ftp2-zlb.vips.scl3.mozilla.com] [ftp.dyndect.mozilla.net] (Linux) TCP 21	
Server: 74.125.225.101 [clients1.google.com] [clients1.google.com] TCP 80	
Server: 74.125.225.180 [www.google.com] TCP 80	
Server: 209.128.72.218 [zeemud.org] [mud.zeemud.org] (Linux) TCP 4000	
Host Details	
Queried DNS names: clients1.google.com,ftp.download.com,ftp.mozilla.org,mud.zeemud.org	
Web Browser User-Agent 1: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)	
Device Category: Windows	
Accept-Language: en-us	

4.2.3 Files Screen Grab

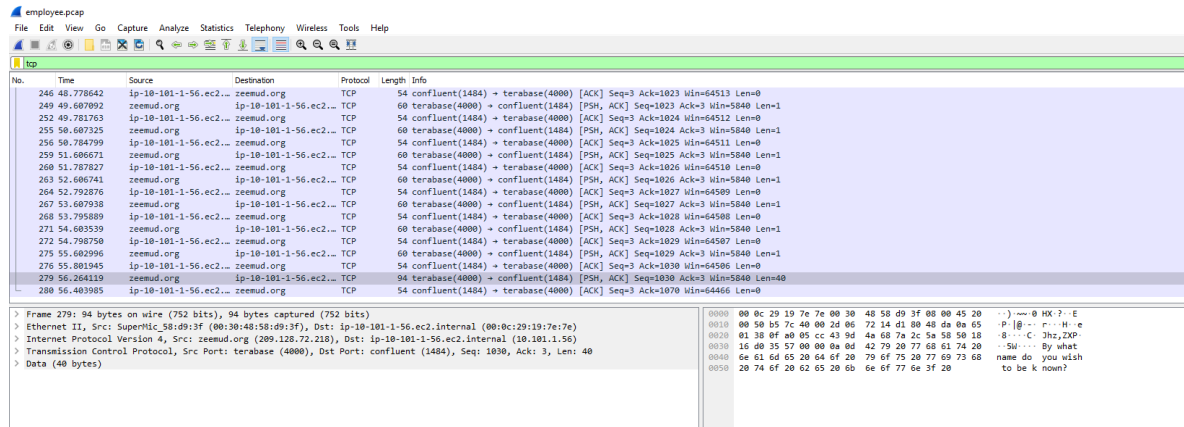
Hosts (18) Files (3) Images Messages Credentials (3) Sessions (9) DNS (19) Parameters (51) Keywords Anomalies									
Filter keyword:									
Filename	Extension	Size	Source host	S. port	Destination host	D. port	Protocol	Timestamp	
search.4AC26891[3].html	html	46 408 B	74.125.225.180 [www.google.com]	TCP 80	10.101.1.56 (Windows)	TCP 1473	HttpGetChunked	2012-08-15 01:06:28 UTC	
README[6]		528 B	63.245.215.56 [ftp2-zlb.vips.scl3.mozilla.com] [ftp.dyndect.m...	TCP 7011	10.101.1.56 (Windows)	TCP 5001	FTP	2012-08-15 01:06:44 UTC	
README[7]		528 B	63.245.215.56 [ftp2-zlb.vips.scl3.mozilla.com] [ftp.dyndect.m...	TCP 7011	10.101.1.56 (Windows)	TCP 5002	FTP	2012-08-15 01:06:51 UTC	

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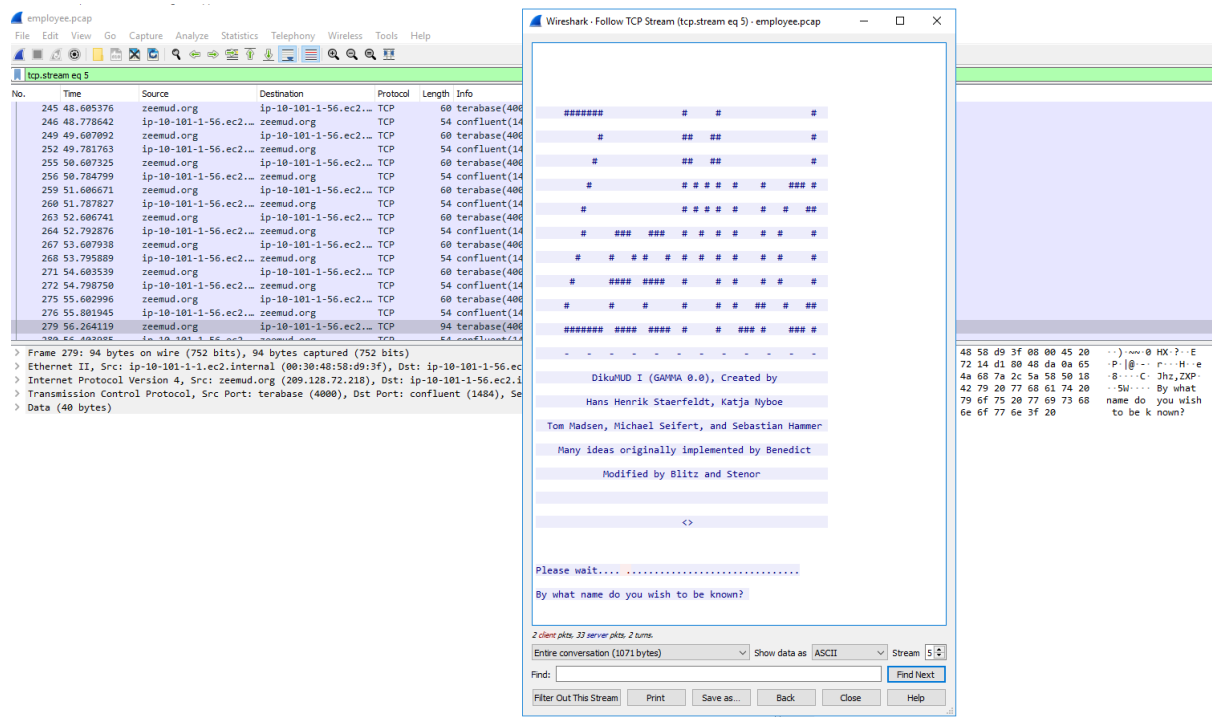
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4.3 WireShark Screen Grabs

4.3.1 TCP Filter Screen Grab



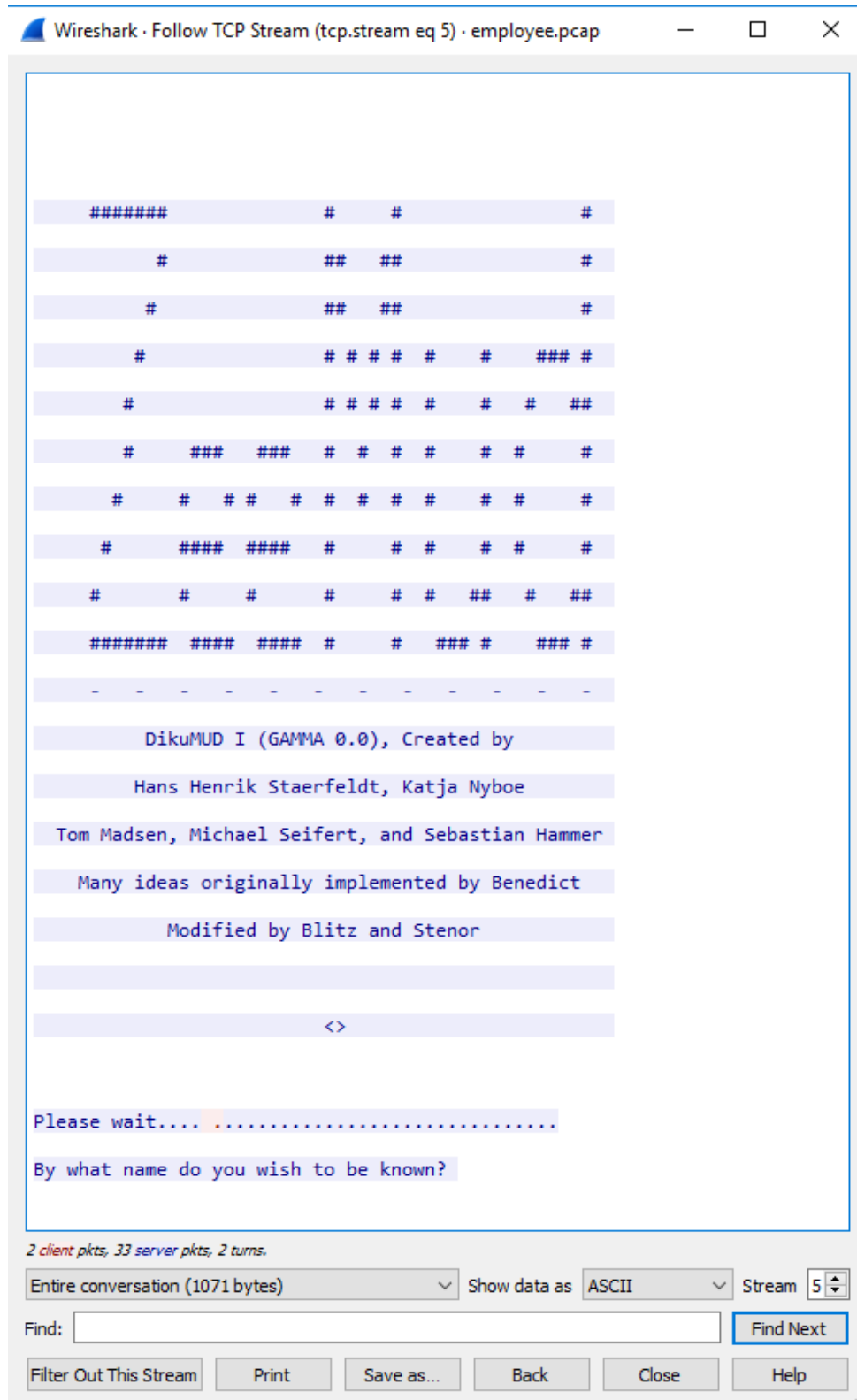
4.3.2 Follow TCP Stream



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4.3.3 Weird Text Screen Grab



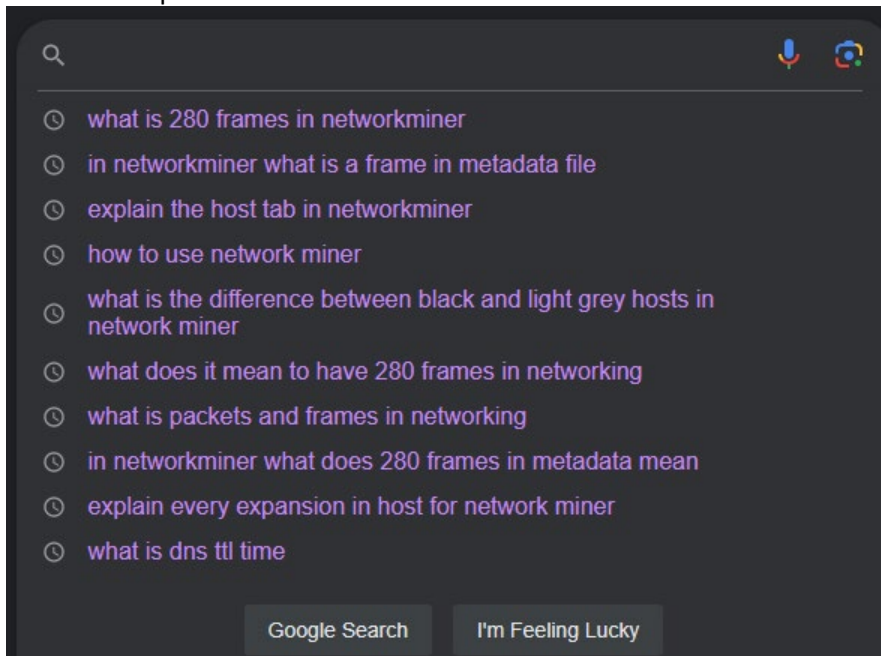
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5 Self-Directed Learning

I found it frustrating to try to find out more about NetworkMiner. Even the user manual provided by NetreseC is minimal at best.

Here's a sample of what I asked:



I see NetworkMiner providing a lot of information that can help tell a story, but I would like to be able to see how the dots connect...the one thing I did learn...repeatedly online, and now I feel quite proficient at, is: How to download it...and the name of the tabs...

I was able to find out some information, but it still is very lacking in substance. Most tutorials show you how to navigate around, but not what you're looking at and how it applies to your investigation.

Thanks to my Mentor, I could see a bigger picture, and maybe I'm delving too deeply into the data...but I figure all the information provided is there to tell and direct you towards an informed conclusion...or at least where to go to next.

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5.1 Questions:

1. On the Host Tab what am I seeing:
 - a. Incoming Sessions
 - b. Outgoing sessions
 - c. Every expandable...what are they telling me, what is the information being given, and how to use it.
2. Why do some operations show up in parameters and others don't.
 - a. Example: You can see google ops and Mozilla ops, but not Zee Mud.
3. Why no images?
 - a. It is reported that the employee downloaded two images as stated in Files and elsewhere, but they aren't in the Images tab.
 - b. When would images be captured?
4. In general, how do all the tabs interact with each other to tell the whole story...what is the best path to follow when doing an investigation?
5. How to see malware on NetworkMiner?
 - a. At this time, I cannot completely tell because of various reasons supplied online.
 - i. The first example video online does show that the malware would have pinged an antivirus warning on the VM. There was no warning, so possibly no malware or virus.
 - ii. The second example requires WireShark to work with NetworkMiner. We're not there yet.
 - iii. The third example shows NetworkMiner working with WireShark and MalMiner...both of which we haven't started yet.
 - iv. The fourth example says you need Network Miner Pro to see Malware redirects. We are not using that.
6. How do you get the second row of tabs Name, Format, VT Detection, Yara, Suspicion, PEID, Local Report VT Report and their uses on NetworkMiner?

